

An endowment economy

1) Environment

- discrete time $t = 0, 1, 2, \dots, \tau$
- one good, perishable

τ
↑
odd number

Households

two types $e \in \{A, B\}$, get endowments y_t^e

$e \backslash t$	0	1	2	3	4	...	τ
A	1	0	1	0	1	...	0
B	0	1	0	1	0	...	1

- n^A type A Hts, n^B type B Hts "masses of agents"

Preferences

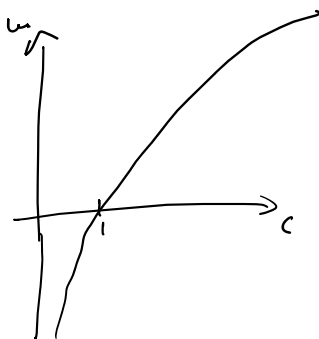
$$U_t = \sum_{t=0}^{\tau} \beta^t u(c_t^e)$$

↓
 discount factor $\beta \in (0, 1)$

↑
 consumption

$$u(c) = \ln(c)$$

$$\underline{c_t^e \geq 0} \quad \forall t$$



- households can borrow or save $s_t^e > \text{save}$
 $s_t^e < \text{borrow}$
- face prices $p_t > 0$

$$\underline{u' > 0}$$

time 0

$$\underbrace{p_0 x_0^e}_{\$ \text{ cons.}} + \underbrace{s_0^e}_{\text{savings}} \leq \underbrace{p_0 y_0^e}_{\$ \text{ income}}$$

time $t > 0$

$$p_t c_t^e + s_t^e \leq p_t y_t^e + \underbrace{s_{t+1}^e (1+i_{t+1})}_{\text{nominal interest rate}}$$

Interpretation:

income

$$p_t c_t + s_t = \underbrace{1 + i_t}_{\text{income}} \underbrace{s_{t-1}}_{\text{income}}$$

Digression:

Suppose inequality was strict " $<$ " for some $\tilde{t}$ at solution x

$$p_{\tilde{t}} c_{\tilde{t}}^x + s_{\tilde{t}}^x < p_{\tilde{t}} y_{\tilde{t}} + s_{\tilde{t}-1}^x (1 + i_{\tilde{t}+1})$$

$$U_t^x = \sum_{t=0}^{\infty} \beta^t u(c_t^x)$$

$$U_t^{x*} = \sum_{\substack{t=0 \\ t \neq \tilde{t}}}^{\infty} \beta^t u(c_t^x) + \beta^{\tilde{t}} u(c_{\tilde{t}}^{x*}) > U_t^x$$

where

$$c_{\tilde{t}}^{x*} = p_{\tilde{t}} y_{\tilde{t}} + s_{\tilde{t}-1}^x (1 + i_{\tilde{t}+1}) - s_{\tilde{t}}^x > c_{\tilde{t}}^x$$

No Ponzi Game condition

$$s_T \geq 0$$

$$T \rightarrow \infty$$

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^T s_t \geq 0$$

"the UU cannot die in debt"

$$t=0 \quad c_0^e + b_0^e \leq y_0^e \quad \text{where } b_0^e = \frac{[s_0^e]}{[P_0]} = \frac{\$}{\text{market}}$$

$$t > 0 \quad c_t^e + b_t^e \leq (1 + i_{t-1}) \cdot \frac{P_{t-1}}{P_t} b_{t-1}^e + y_t^e$$

$$c_t^e + b_t^e \leq y_t^e + b_{t+1}^e (1 + i_{t+1})$$

$$\pi_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

$$1 + \pi_t = \frac{P_t}{P_{t-1}}$$

$$\frac{1 + i_t}{1 + \pi_{t+1}} = 1 + i_{t+1}$$

$$i_t = \sigma_t + \pi_{t+1} \quad \text{Fisher equation}$$

Household problem:

Taking interest rates $\{r_t\}_{t=0}^{T-1}$ as given, households solve the following problem:

$$T, t, e$$

$$\max \sum \beta^t u(c_t)$$

solve the following problem.

$$\max_{\{c_t^e, b_t^e\}_{t=0}^T} \sum_{t=0}^T \beta^t \ln(c_t^e)$$

$$c_t^e + b_t^e \leq y_t^e + (1+r_{t-1})b_{t-1}^e$$

$$\text{with } b_{-1}^e = 0$$

$$\text{and } b_T^e \geq 0$$

$$(c_t^e \geq 0)$$

$$\max \sum \beta^t \ln(c_t)$$

s.t.

budget cons.

+ upg. cost

$$+ r(b_t)$$

+/-

Equilibrium :

A competitive equilibrium in this economy is a set of prices $\{r_t\}_{t=0}^{T-1}$ and a set of allocations $\{c_t^A, c_t^B, b_t^A, b_t^B\}_{t=0}^T$ such that

specifies market structure

1) Taking $\{r_t\}_{t=0}^{T-1}$ as given, the allocations $\{c_t^A, c_t^B, b_t^A, b_t^B\}$ solve the hts' optimization problems.

2) The goods market clears for $t=0, \dots, T$

$$\text{demand} \quad n^A \cdot c_t^A + n^B \cdot c_t^B = \text{supply} \quad n^A y_t^A + n^B y_t^B$$

$$Y = C + \cancel{I} + \cancel{G} + \cancel{X}$$

$$\sum^e p(\text{demand} - \text{supply}) \leq 0 \quad \sum Y$$

and the market for assets clears

$$n^A b_t^A + n^B b_t^B = 0$$

Model Solution

Solving the hts' problem

$$\max_{\{c_t^e, b_t^e\}_{t=0}^T} \sum_{t=0}^T \beta^t \ln(c_t^e)$$

s.t.

$$y_t^e + (1+r_{t-1})b_{t-1}^e - c_t^e - b_t^e \geq 0$$

$$b_T^e \geq 0$$

$$\beta = \frac{1}{1+\rho}$$

rate of discount time preference

$$[\rho] = \frac{\text{utils}}{\text{goods}}$$

$$b_T^e \geq 0 \quad [\lambda] = \frac{\text{utils}}{\text{goods}}$$

Current-value Lagrangian

$$\mathcal{L}^e = \sum_{t=0}^T \beta^t \left[\ln c_t^e + \lambda_t^e (y_t^e + (1+r_{t-1})b_{t-1}^e - c_t^e - b_t^e) + \mu_T^e b_T^e \right]$$

$$\lambda_t^{e, PV} = \beta^t \lambda_t^e$$

$$\mathcal{L}^{PV} = \sum_{t=0}^T \beta \ln c_t^e + \sum_{t=0}^T \lambda_t^{e, PV} (y_t^e + (1+r_{t-1})b_{t-1}^e - c_t^e - b_t^e) + \mu_T^e b_T^e$$

F.O.C. $\frac{\partial \mathcal{L}^e}{\partial c_t^e} = 0 \Leftrightarrow \beta^t \left(\frac{1}{c_t^e} - \lambda_t^e \right) = 0$

$$\boxed{\frac{1}{c_t^e} = \lambda_t^e} \quad e=A, B \quad (1)$$

$t=0, 1, \dots, T$

$$\frac{\partial \mathcal{L}^e}{\partial b_t^e} = 0 \Leftrightarrow -\beta^t \lambda_t^e + \beta^{t+1} \lambda_{t+1}^e (1+r_t) = 0$$

$$\boxed{\lambda_t^e = \beta (1+r_t) \lambda_{t+1}^e} \quad (2)$$

$t=0, 1, \dots, T-1$

$t=T$

$$\mu_T^e - \lambda_T^e \beta^T = 0$$

$$\boxed{\mu_T^e = \beta^T \lambda_T^e} \quad (3)$$

1. constraints hold

$$y_t^e + (1+r_{t-1})b_{t-1}^e - c_t^e - b_t^e \geq 0 \quad \forall t=0, \dots, T \quad (4)$$

$$b_T^e \geq 0 \quad (5)$$

2. complementary slackness

$$\lambda_t^e (y_t^e + (1+r_{t-1})b_{t-1}^e - c_t^e - b_t^e) = 0 \quad (6)$$

$$\mu_T^e b_T^e = 0 \quad (7)$$

3. non-negativity

$$\lambda_t^e \geq 0 \quad t=0, \dots, T \quad (8)$$

non-negativity

$$\lambda_t^e \geq 0 \quad t=0, \dots, T \quad (8)$$

$$\mu_T^e \geq 0 \quad (9)$$

$q^e > 0 \Rightarrow \lambda_t > 0 \Rightarrow (8) \text{ satisfied}$
 $(4) \text{ holds with}$

flow / per-period budget constraint $\Rightarrow \boxed{c_t^e + b_t^e = y_t^e + (1+r_{t-1})b_{t-1}^e} \quad (*)$
 $(6) \text{ holds}$

$\Rightarrow \mu_T^e = \beta^T \lambda_T^e > 0 \Rightarrow (9) \text{ satisfied}$
 $\Rightarrow \boxed{b_T^e = 0} \Rightarrow (5) \text{ satisfied}$

$\Rightarrow \boxed{\frac{1}{c_t^e} = \beta(1+r_t) \cdot \frac{1}{c_{t+1}^e}} \quad \text{Euler equation} \quad (**)$

$\Rightarrow b_{-1}^A = b_{-1}^B = 0$

$\Rightarrow b_T^A = b_T^B = 0 \quad \text{transversality condition (TVC)}$

boils down to 2 equations $(*)$, $(**)$
 and two boundary conditions $b_{-1}^e = 0$, $b_T^e = 0$

The intertemporal budget constraint

$$b_t^e = y_t^e - c_t^e + b_{t+1}^e(1+r_{t+1})$$

$\rightarrow$ solution
 consumption
 fun. $c_t = f(\{y_t, r_t\}_{t=0}^T)$

$$\underbrace{\sum_{t=0}^T \frac{1}{\prod_{\tau=0}^{t-1} (1+r_\tau)}}_{\text{life-time consumption}} c_t = \underbrace{\sum_{t=0}^T \frac{1}{\prod_{\tau=0}^{t-1} (1+r_\tau)} y_t}_{\text{life-time income}}$$

Some properties of the equilibrium from $(**)$

$\cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot$
 $c_{t+1}^e = \beta(1+r_t) c_t^e$

Some properties of the equilibrium from (88)

$$u^A y_t^A + u^B y_t^B = u^A c_t^A + u^B c_t^B$$

$$= u^A \frac{c_{t+1}^A}{\beta(1+r_t)} + u^B \frac{c_{t+1}^B}{\beta(1+r_t)}$$

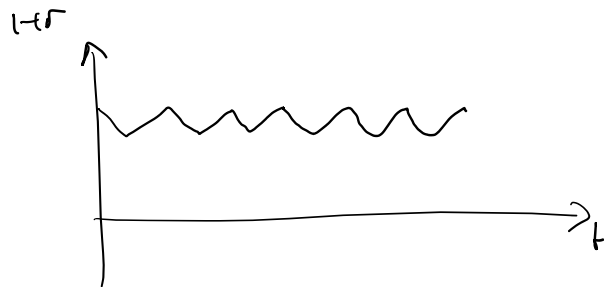
$$c_{t+1}^e = \beta(1+r_t) c_t^e$$

$$= \frac{1}{\beta(1+r_t)} \left(u^A c_{t+1}^A + u^B c_{t+1}^B \right)$$

$$= \frac{1}{\beta(1+r_t)} \left(u^A y_{t+1}^A + u^B y_{t+1}^B \right)$$

$$1+r_t = \frac{1}{\beta} \frac{u^A y_{t+1}^A + u^B y_{t+1}^B}{u^A y_t^A + u^B y_t^B}$$

$$1+r_t = \begin{cases} \frac{1}{\beta} \frac{u^B}{u^A} & \text{if } t \text{ even} \\ \frac{1}{\beta} \frac{u^A}{u^B} & \text{if } t \text{ odd} \end{cases}$$



Extensions

Infinite horizon economies

$$T \rightarrow \infty$$

Consider the following borrowing scheme:

$$t=0 \quad \text{borrow } b_0 = -x, \quad x > 0$$

$$t=1 \quad \text{borrow } b_1 = -x(1+r_0)$$

$$t=2 \quad \text{" } b_2 = -x(1+r_0)(1+r_1)$$

T-1

this scheme
violates the

$$t=2 \quad b_2 = -x(1+r_0)(1+r_1)$$

$$t=T \quad \text{borrow } b^T = -x \prod_{\tau=0}^{T-1} (1+r_\tau)$$

This scheme violates the upg condition!

no Ponzi-game condition:

$$\lim_{T \rightarrow \infty} \frac{b_T}{\prod_{\tau=0}^{T-1} (1+r_\tau)} \geq 0$$

PV multiplier

The corresponding TUC could replace with

$$\lim_{T \rightarrow \infty} \beta^T \lambda_T b_T = 0$$

Use $\lambda_t = \frac{1}{\beta(1+r_{t+1})} \lambda_{t+1}$

$$= \dots = \frac{1}{\beta^T \prod_{\tau=0}^{T-1} (1+r_\tau)} \lambda_0$$

at infinity ⁽¹⁾
the present ⁽²⁾
value of asset ⁽³⁾
holdings is zero ⁽⁴⁾

$$\Rightarrow \lim_{T \rightarrow \infty} \beta^T \frac{1}{\beta^T \prod_{\tau=0}^{T-1} (1+r_\tau)} \lambda_0 b_T = 0 \quad \left. \vphantom{\lim_{T \rightarrow \infty}} \right\} : \lambda_0$$

$\Rightarrow$ no Ponzi-game condition follows with "="

Uncertainty

$$y_t = \bar{y} + \varepsilon_t$$

$$\varepsilon_t \stackrel{iid}{\sim} G(\varepsilon_t)$$

$$E[\varepsilon_t] = 0 \quad \forall t$$

It becomes

$$\max_{\{c_{t+j}, b_{t+j}\}_{j=0}^{\infty}} E_t \sum_{j=0}^{\infty} \beta^j \ln(c_{t+j})$$

s.t.

$$(i) \quad c_{t+j} + b_{t+j} \leq y_{t+j} + b_{t+j-1} (1+r_{t+j-1})$$

(ii) nPg condition

$$\text{and } (1+r_{t-1})b_{t-1} = 0$$

$$\mathcal{L} = E_t \sum_{j=0}^{\infty} \beta^j \left(\ln c_{t+j} + \lambda_{t+j} \left(y_{t+j} + (1+r_{t+j-1})b_{t+j-1} - c_{t+j} - b_{t+j} \right) \right)$$

current-value multiplier

$$\frac{\partial \mathcal{L}}{\partial c_t} = 0 \quad \Leftrightarrow \quad E_t \left[\frac{1}{c_t} - \lambda_t \right] = 0$$

$$E_t[\cdot] = E[\cdot | \{y_{t+j}, r_{t+j}, b_{t+j-1}\}_{j=0, \dots, \infty}]$$

$$\frac{1}{c_t} = \lambda_t$$

TVC
 $\lim_{T \rightarrow \infty} \beta^T \lambda_T b_T = 0$
 PV version
 $\lim_{T \rightarrow \infty} \lambda_T b_T = 0$

$$\frac{\partial \mathcal{L}}{\partial b_t} = 0$$

$$E_t \left[\beta (1+r_t) \lambda_{t+1} - \lambda_t \right] = 0$$

$$\Rightarrow \lambda_t = \beta (1+r_t) E_t [\lambda_{t+1}]$$

$$\frac{1}{c_t} = \beta (1+r_t) E_t \left[\frac{1}{c_{t+1}} \right]$$

$$E_t \left[\frac{1}{c_{t+1}} \right] = \int \frac{1}{c_{t+1}} g(\varepsilon_{t+1}) d\varepsilon_{t+1}$$